



Personalized Levodopa Prescription Using Hybrid Reinforcement Learning in Parkinson's Disease

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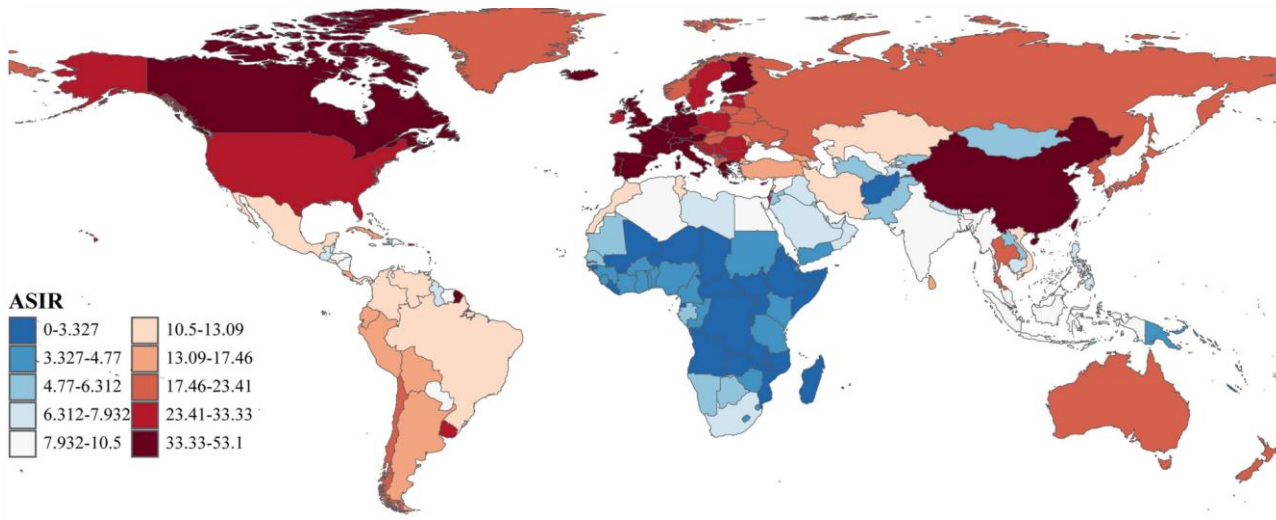
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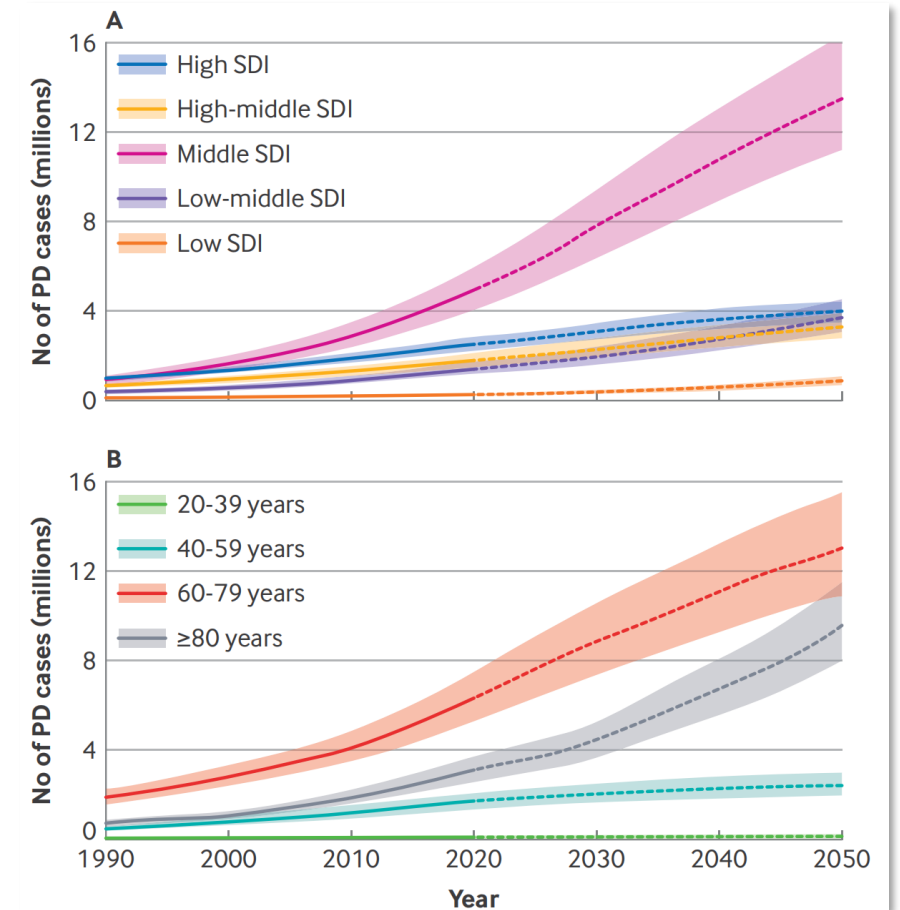
1. Prevalence of Parkinson's Disease (PD)

About **11.77 million** people worldwide were living with PD in 2021, representing a **274% increase** from 3.15 million in 1990



The Age-Standardized Prevalence Rate of PD in 2021 (Luo et al., 2025)

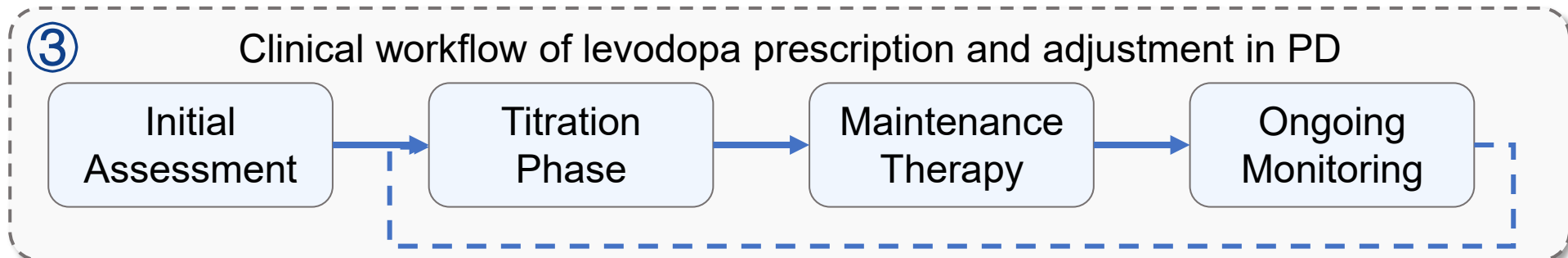
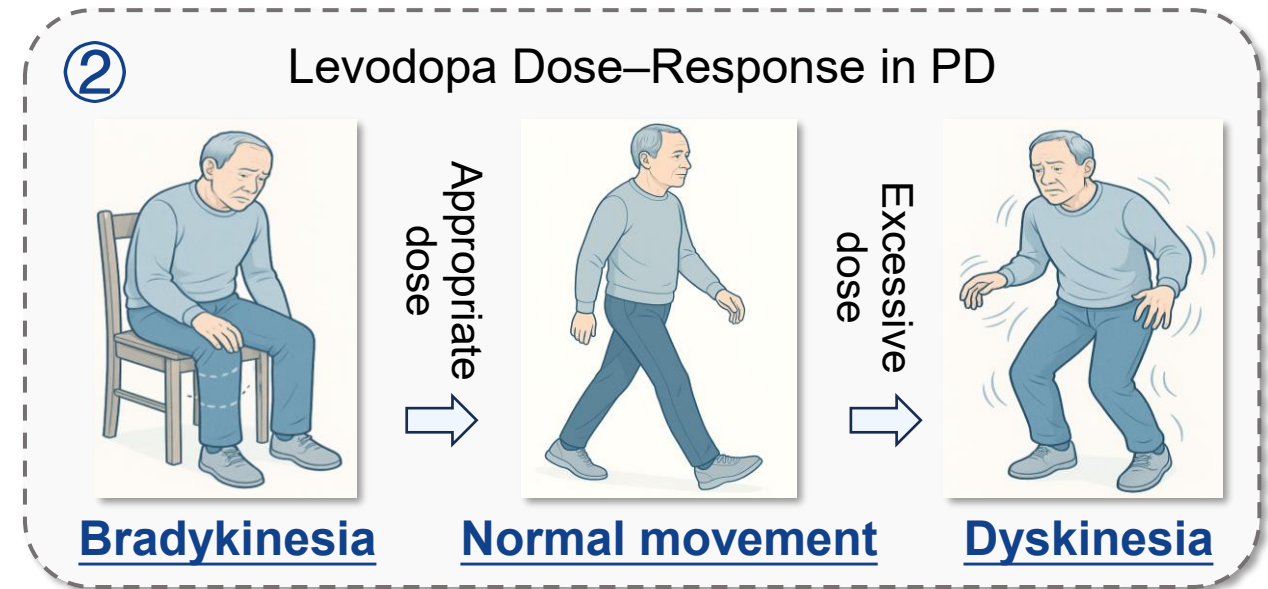
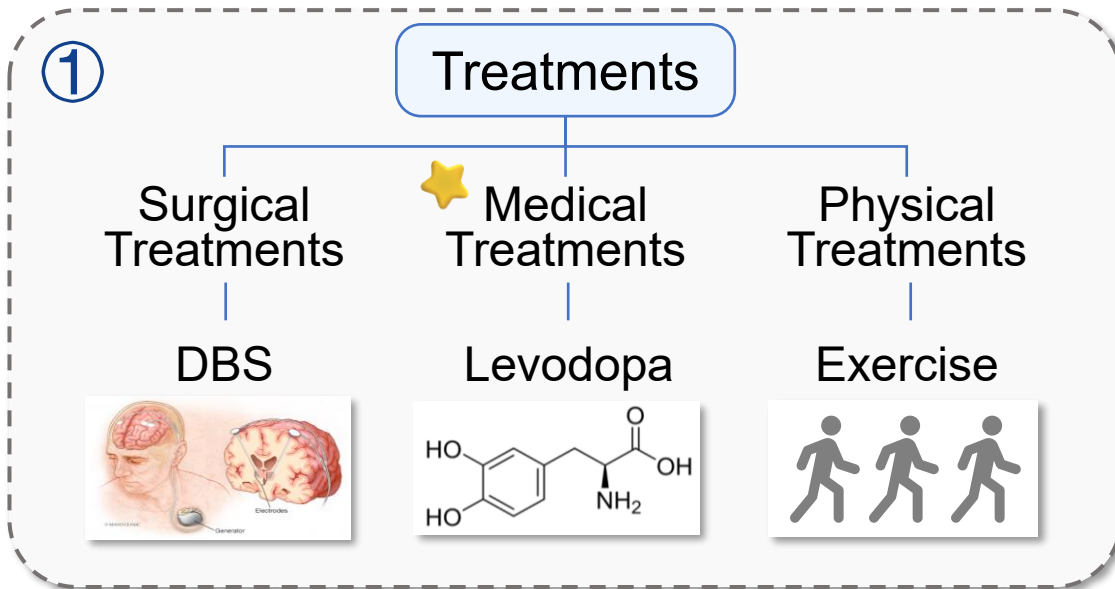
The prevalence is further projected to reach **25.2 million** by 2050, representing a **114% increase** from 2021



Estimated trends of numbers of PD cases (Su et al., 2025)

2. Parkinson's Disease and Its Treatment

Parkinson's Disease represents a fast-growing neurodegenerative condition. Current criteria define Parkinson's disease as the presence of bradykinesia combined with either rest tremor, rigidity, or both. (Bloem et al., 2021)



3. Motivation & Method Overview

Clinical Motivation

I. Automated Dosing Support

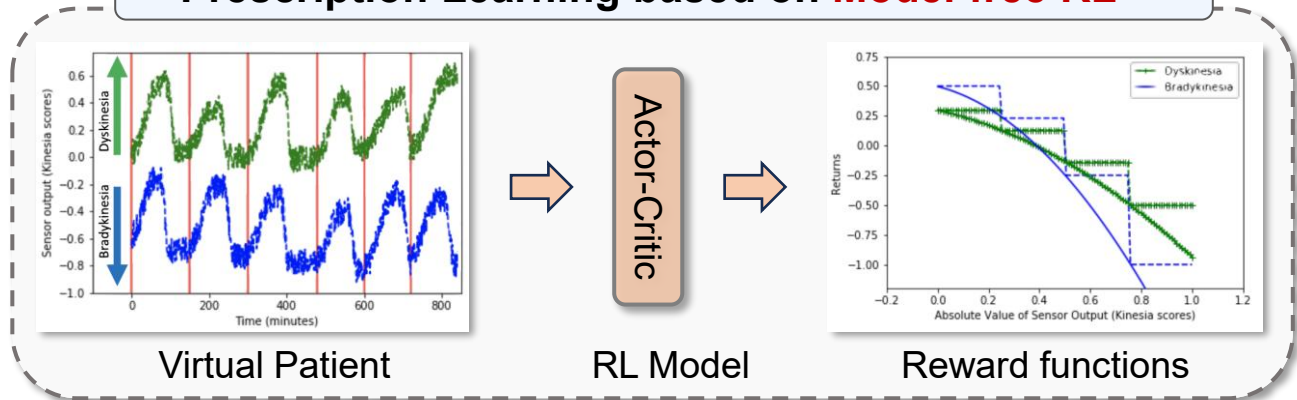
Automated algorithms for personalized levodopa dosing can **streamline clinical decision-making** and **reduce clinician workload**.

II. Dynamic Dosing Needs

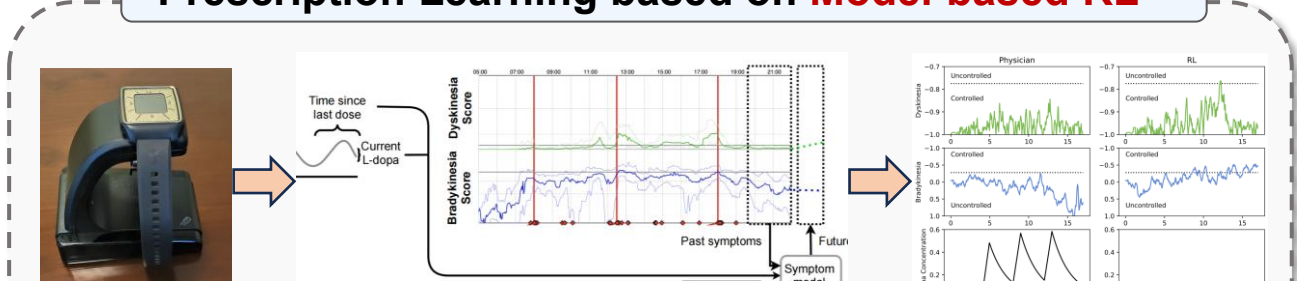
Real-time monitoring and **adaptive dosing** are needed to manage fluctuating levodopa levels effectively.

Method Overview

Prescription Learning based on **Model-free RL**^[1]



Prescription Learning based on **Model-based RL**^[2]



III. Model-free methods pose **safety risks in early clinical deployment, whereas model-based methods suffer from **model bias**, limiting their **reliability** for real-world dosing decisions.**

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1. Framework for Personalized Levodopa Prescription

The **overall framework** begins with the ① construction of a virtual patient environment incorporating pharmacokinetic mechanisms. An ② initial dosing policy is then generated using model-based RL. This policy is subsequently ③ fine-tuned and optimized through model-free RL via simulated interactions.

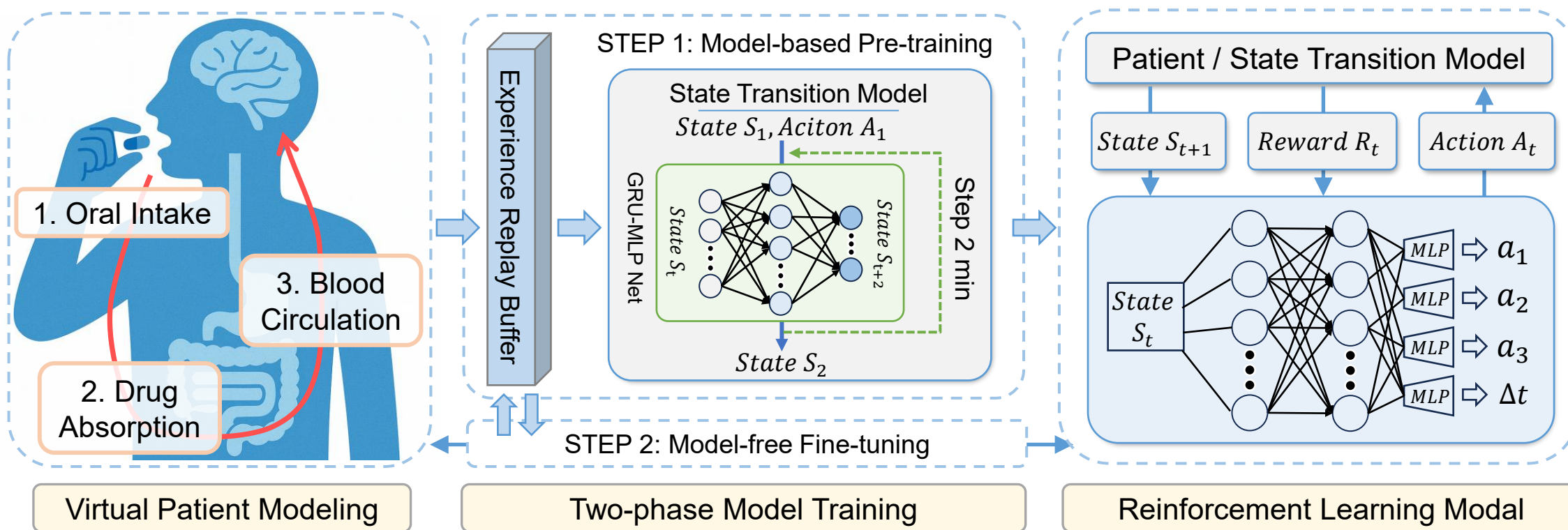


Figure 1. Overall framework of the proposed two-phase RL approach for personalized levodopa prescription

2. Construction of the Virtual Patient Environment

(1) Pharmacokinetic Modeling of Levodopa

- Build a conversion model to unify different levodopa formulations into a standardized **Levodopa Equivalent Dose (LED)**:

$$L(t) = \begin{cases} \frac{\rho_{peak}}{t_{peak}}(t - t_{past})n, & t - t_{past} < t_{peak}; \\ n\rho_{peak}e^{-\theta(t-t_{past}-t_{peak})}, & t - t_{past} > t_{peak}; \end{cases}$$

$L(t)$ - LED t_{peak} - time to peak t_{past} - time since previous dosing ρ_{past} - half-time

- Select three **typical L-dopa formulations** (IR, CR, and Rytary):

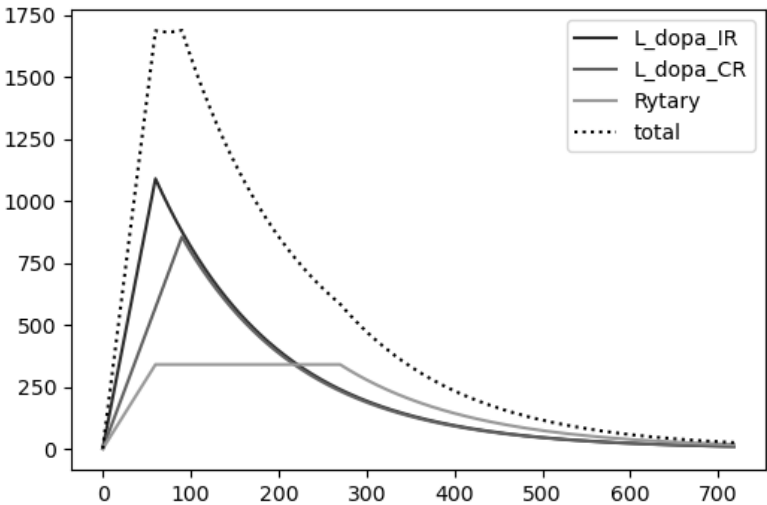
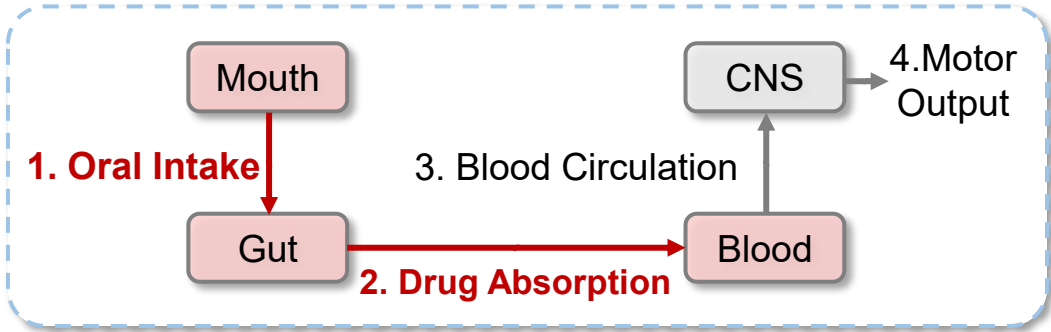
Table 1. Pharmacokinetics of L-dopa formulations (Baucum et al., 2023)

Drug Type	Time to Peak (min)	Peak Concentration (ng/(mL·mg))	Half-life (min)
L-dopa IR	60	10.9	96
L-dopa CR	90	8.55	96
Rytary	60	3.4	114

*Rytary sustains peak concentration for approximately 3.5 hour.

- **Weighted summation** to represent combined dosing response:

$$L^{total}(t) = \alpha_1 L^{IR}(t) + \alpha_2 L^{CR}(t) + \alpha_3 L^{Ry}(t)$$



Simulated plasma levodopa-equivalent concentration–time curve after a 100 mg dose (Baucum et al., 2023)

2. Construction of the Virtual Patient Environment

(2) Modeling of Blood-to-CNS Transport Mechanism

- Characterize the dynamic transport of orally administered levodopa **from the bloodstream into the central nervous system (CNS)**:

$$\begin{cases} n_{\text{CNS}}(t+1) = n_{\text{CNS}}(t) + \sum_{i=0}^t \frac{\Delta n(i)}{2^{\frac{\alpha}{2}} \Gamma(\frac{\alpha}{2})} x^{\frac{\alpha}{2}-1} e^{-\frac{t-i}{2}} + \epsilon \\ \Delta n(t) = n_{\text{CNS}}(t) - n_{\text{blood}}(t) \end{cases}$$

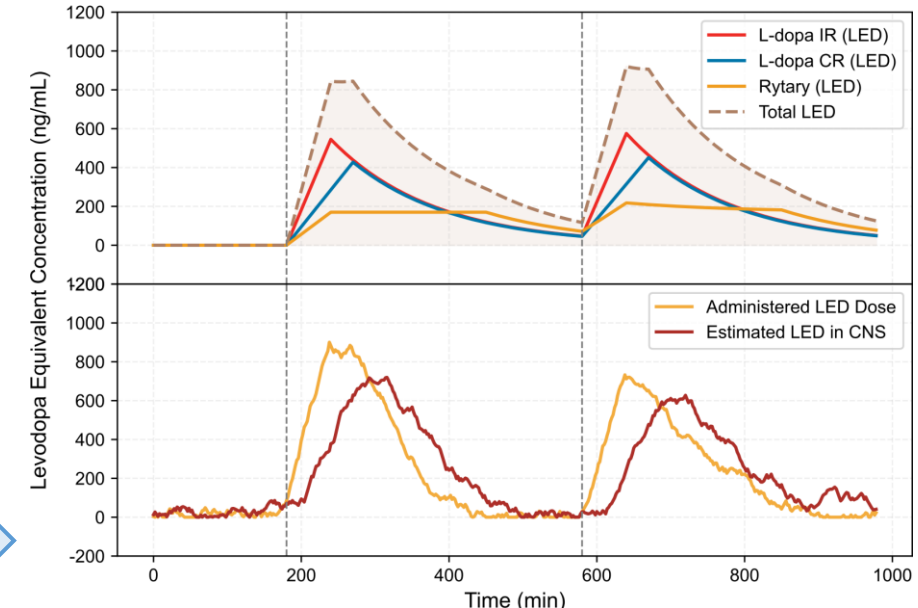
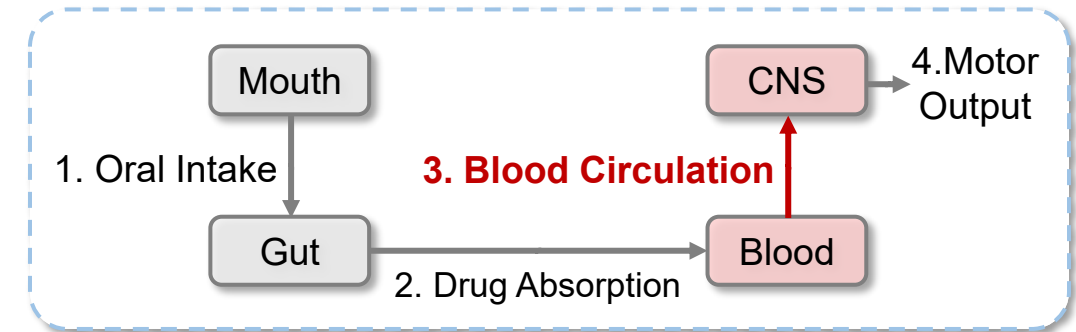
$n_{\text{CNS}}(t)$ - Concentration of levodopa in CNS;

$n_{\text{blood}}(t)$ - Levodopa concentration in bloodstream;

$\sum_{i=0}^t \frac{\Delta n(i)}{2^{\frac{\alpha}{2}} \Gamma(\frac{\alpha}{2})} x^{\frac{\alpha}{2}-1} e^{-\frac{t-i}{2}}$ - Chi-square distribution describes the **time-lag effect** of levodopa transport;

α - Characterizes **the latency of drug transport**, influenced by factors such as **age**, **sex** and **medication history**;

- Construct a virtual **80-year-old female** PD patient with a **5-year disease history** and simulated the absorption and blood-to-CNS transport of 100 mg doses of L-dopa IR, CR, and Rytary.



Pharmacokinetic simulation of three L-dopa formulations, illustrating drug absorption and concentration profiles in the bloodstream and CNS

2. Construction of the Virtual Patient Environment

(3) Modeling of Functional Response in Virtual Patient

- Characterize **motor responses** using **bradykinesia** and **dyskinesia scores** driven by dynamic CNS levodopa levels. (Lower score indicates a better motor state)

$$b_t = b_{t-1} + \Delta \widehat{b}_t + \epsilon_b$$
$$d_t = d_{t-1} + \Delta \widehat{d}_t + \epsilon_d$$

b_t - Bradykinesia score at time t ; d_t - Dyskinesia score at time t ;
 $\Delta \widehat{b}_t, \Delta \widehat{d}_t$ - Symptom change **driven by CNS levodopa concentration**;



$$\Delta b_t = \begin{cases} 1 - \frac{n_{\text{CNS}}}{C_1}, & \text{if } n_{\text{CNS}} < C_1 \\ -1 + \frac{(n_{\text{CNS}} - C_2)^2}{(C_2 - C_1)^2}, & \text{if } C_1 \leq n_{\text{CNS}} < C_2 \\ -1, & \text{if } n_{\text{CNS}} \geq C_2 \end{cases}$$
$$\Delta d_t = \min \left(1, -1 + \frac{n_{\text{CNS}}^2}{C_2^2} \right)$$

$C_1 = 500 \text{ ng/mL}$ - Minimum effective level;
 $C_2 = 2000 \text{ ng/mL}$ - Threshold above which dyskinesia risk increases

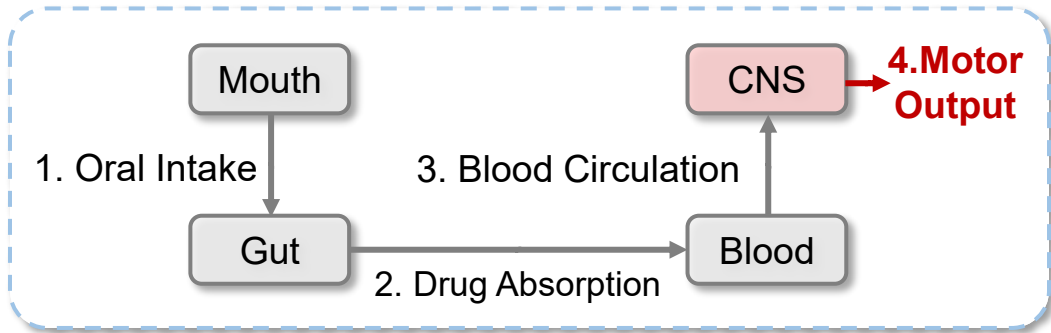


Table 2. Plasma L-dopa Concentration and Clinical Implications (Nelson et al., 1990; Harder et al., 1995)

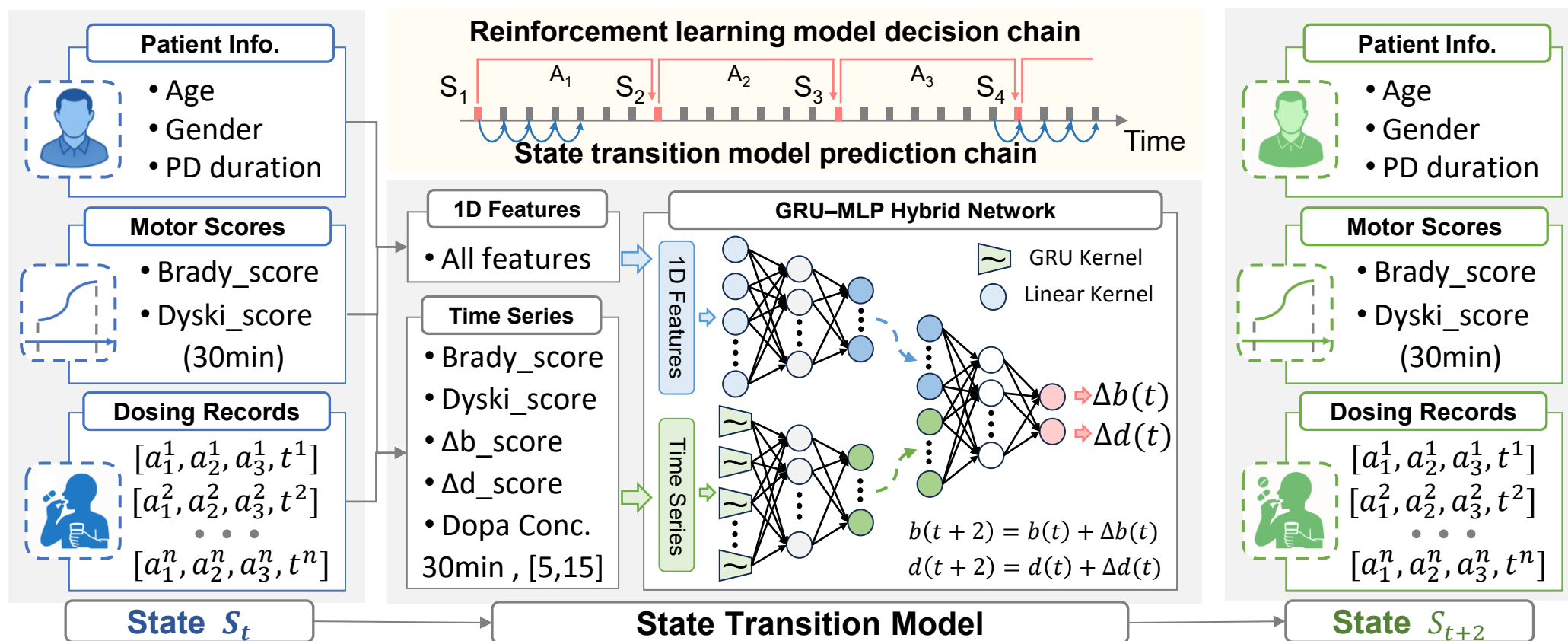
Level	Range (ng/mL)	Clinical Implication
Very low	<500	Below therapeutic threshold; motor symptoms persist (“OFF” state).
Low	500–900	Near threshold; partial symptom control, prone to motor fluctuations.
Optimal	900–2000	Optimal motor control with minimal dyskinesia risk; recommended maintenance range.
High	>2000	Limited additional benefit; increased risk of peak-dose dyskinesia.

3. Construction of the State Transition Model for Virtual Patients

The model learns the temporal progression patterns of patient symptoms and simulates their short-term evolution, using a multi-step transition strategy that preserves **continuous, time-dependent** dynamics beyond single-step prediction.

$$S_t, A_t \rightarrow S_{t+\Delta_1} \rightarrow S_{t+\Delta_2} \rightarrow \dots \rightarrow S_{t+\Delta_t}$$

$$[\Delta b_{t+2}, \Delta d_{t+2}] = f\left(\mathcal{X}_{static}, L_{(t-30:t)}, b_{past}, d_{past}, \Delta b_{past}, \Delta d_{past}\right)$$



4. Hybrid Model-Based and Model-Free Reinforcement Learning Modal

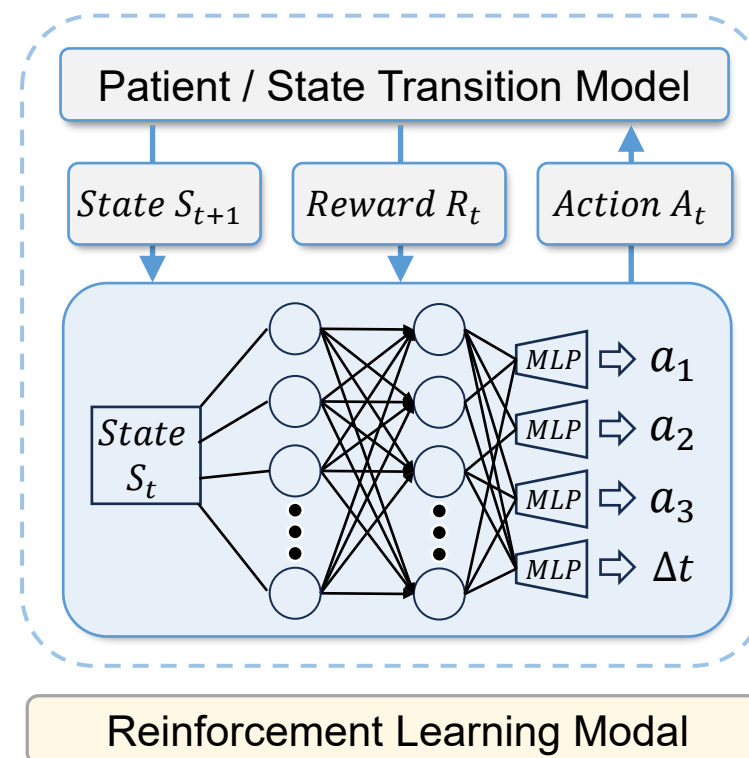
The model provides individualized levodopa prescriptions and ensures safe policy learning by first pre-training on a state transition model and then fine-tuning via secure interaction with a virtual patient.

Phase 1 (Model-based Pre-training): The RL agent first interacts with the state transition model trained on **clinical data**.

→ This pre-training **prevents the agent from generating unsafe or unreliable actions** that could arise from direct interaction at the early learning stage.

Phase 2 (Model-free Fine-tuning): The pre-trained policy is deployed with **virtual patient** environment, where it's refined through **direct interaction**.

→ This phase **enables safe, data-efficient policy improvement** while preserving protection from risky exploration.

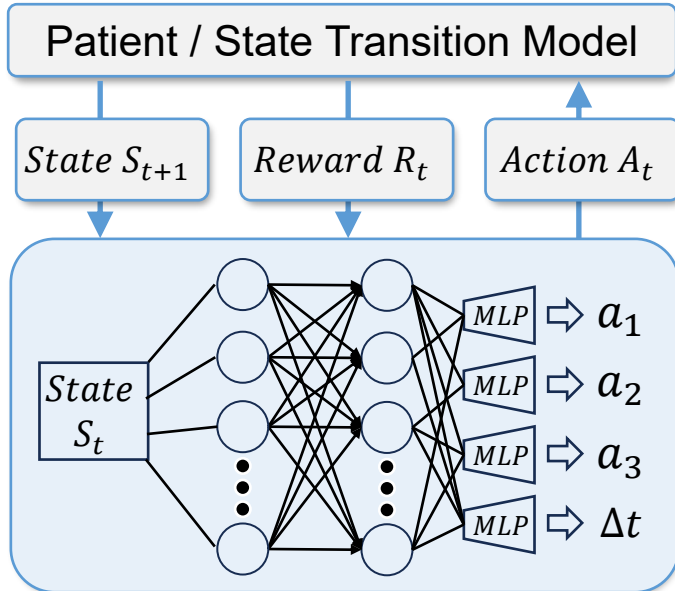


4. Hybrid Model-Based and Model-Free Reinforcement Learning Modal

Action Space

Table 3. Action space of the RL model

Decision Variable	Available Options
L-dopa IR Dosage	0mg, 50mg, 100mg, 150mg, 200mg
L-dopa CR Dosage	0mg, 50mg, 100mg, 150mg, 200mg
Rytary Dosage	0mg, 50mg, 100mg, 150mg, 200mg
Decision Interval	10min, 30min, 60min,...,280min



Reward & Penalty

➤ Penalty for Irregular Medication Intake

$$r_1 = -C_1 \left(\frac{2h - (t - t_p)}{2h} 1_{t-t_p \leq 2h} + \frac{t - t_p - 4h}{4h} 1_{t-t_p > 4h} \right)$$

➤ Reward for Improvement in Motor Function Scores

$$r_2 = C_2^b (1_{b_t \leq b_{t-1}} - 1_{b_t > b_{t-1}}) + C_2^d (1_{d_t \leq d_{t-1}} - 1_{d_t > d_{t-1}})$$

➤ Reward for Maintenance of Motor Function Scores

$$r_3 = -(C_3^b \times b_t + C_3^d \times d_t)$$

Combine the reward components with **weighted sums** and use **time-dependent discounting** to account for **future returns** across uneven decision intervals:

$$G_t = \sum_{k=0}^{T-t} \gamma^{\frac{\Delta t_k}{\tau}} (r_1 + r_2 + r_3)$$

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1. Performance of the State Transition Model

1. Generate a clinical state-transition dataset:

- **100** virtual PD subjects (60–90, ≤10-year medication history).
- **3-day simulation** with a randomized dosing schedule, updated every **2 minutes**.
- **1,437 samples/subject** and **143,700 samples in total**.

2. Define evaluation metrics:

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

3. Training the state transition model:

Adam optimizer; learning rate = 2×10^{-3} ; Batch size = 128

4. Model validation and prediction:

$$MAE = 8.75 \times 10^{-3}, MSE = 1.22 \times 10^{-4}, R^2 = 0.8272$$

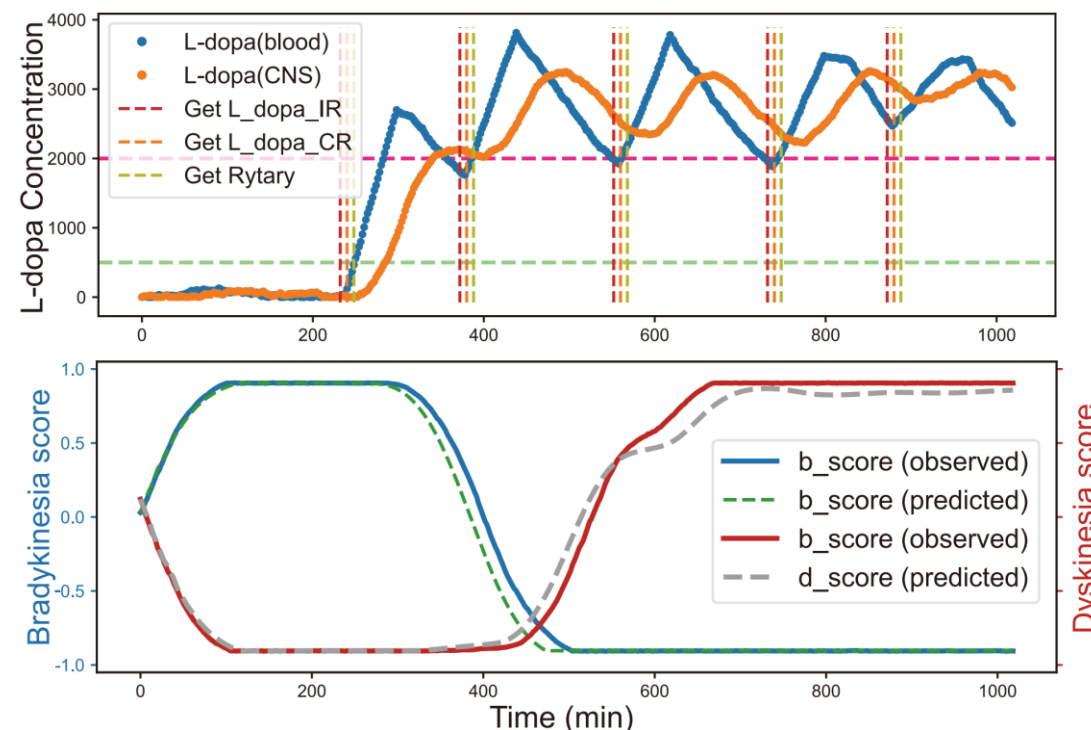


Figure 4. The Simulated pharmacokinetics of L-dopa IR, L-dopa CR, and Rytary in blood and CNS, and comparison of predicted and observed motor symptom scores

2. Performance of the Reinforcement Learning Model

1. Two-phase training strategy

- **Model-based phase:** 1,000 iterations on randomized virtual patients → reward increased from 67.08 → 105.46.
- **Model-free phase:** 200 direct interactions with the virtual patient → reward increased further to 117.70.

2. Model validation

- Applied the learned policy to an 80-year-old female with 5-year medication history.

3. Training outcomes

- Under the policy, L-dopa concentrations in blood and CNS rise promptly after dosing and stabilize within a narrow fluctuation range.
- The bradykinesia score rapidly decreases after dosing and remains low. The dyskinesia score stays minimal throughout the dosing cycles.

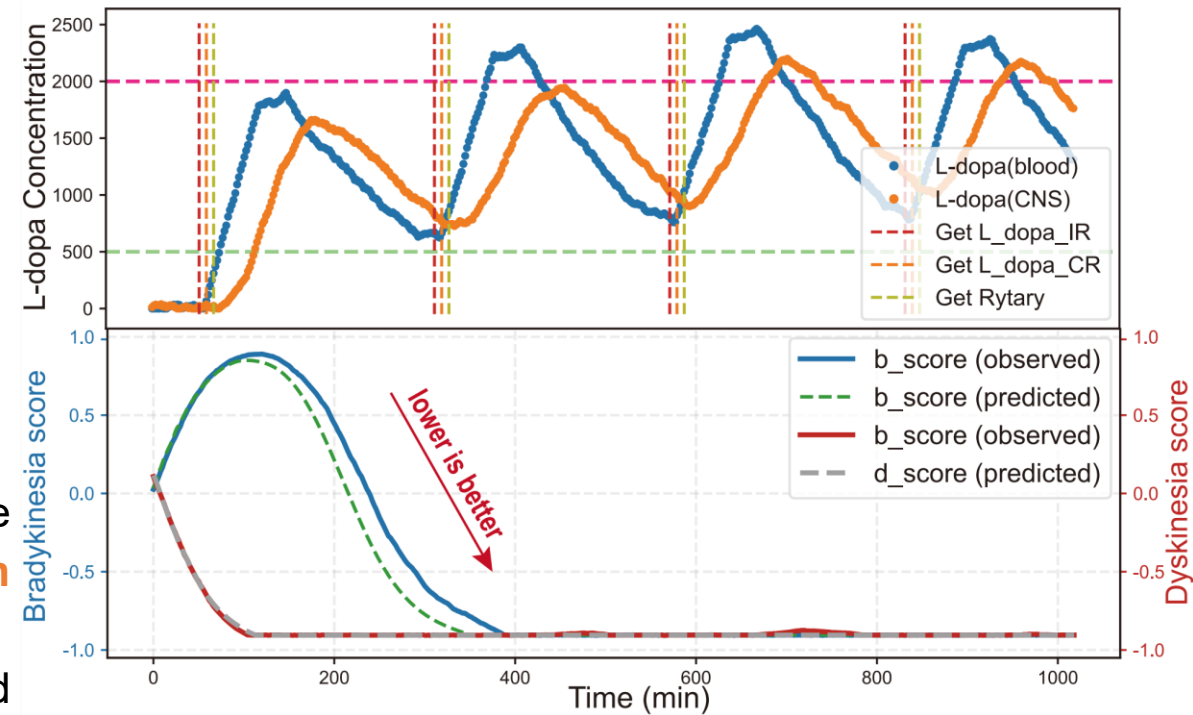


Figure 5. Simulated L-dopa concentration and corresponding bradykinesia and dyskinesia scores under the learned RL policy

The method achieves **stable symptom control** while **keeping dyskinesia at minimal levels**.

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Future Work

1. Transition from virtual simulations to real-world validation

We will integrate wearable sensors and motor-function assessments to validate the method beyond virtual patients and move toward real-world deployment.

2. Incorporating clinically used L-dopa formulations

Beyond the three representative levodopa types currently modeled, future work will expand to include specific formulations used in clinical practice.

3. Integrating non-pharmacological interventions

We will extend the framework to incorporate interventions such as transcranial electrical stimulation and rehabilitation robotics, enabling the generation of comprehensive rehabilitation prescriptions.



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Thank you for your time and attention.

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